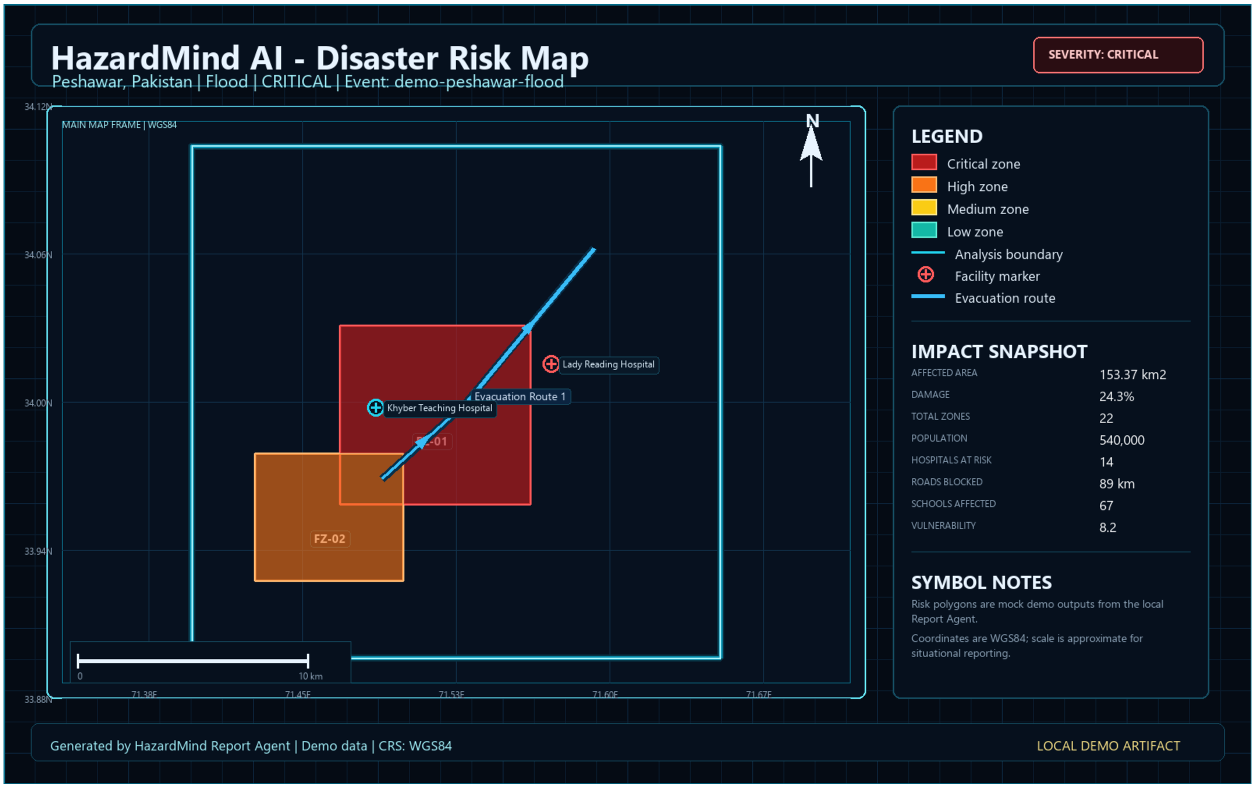
HazardMind AI Executive Disaster Report

Event ID	demo-peshawar-flood
Location	Peshawar, Pakistan
Hazard Type	Flood
Overall Severity	CRITICAL
Satellite Type	sentinel-1
Satellite Reason	cloud_cover_above_30_percent_sar_selected

Operational Statistics

Affected Area	153.37 km2
Damage Percent	24.3%
Total Zones	22
Population Affected	540,000
Hospitals at Risk	14
Roads Blocked	89 km
Schools Affected	67
Vulnerability Score	8.2

Generated Risk Map



Executive Summary

Peshawar faces a CRITICAL flood emergency with approximately 540,000 people exposed across 153.37 km2 of inundation in 22 zones, confirmed by SAR imagery at high confidence (0.91) despite 42% cloud cover. The crisis threatens 14 hospitals, blocks 89 km of roads, and affects 67 schools, with a vulnerability score of 8.2/10. A severe operational bottleneck exists as only one evacuation route currently serves the entire affected population, while deep-water zones FZ-01 and FZ-02 endanger access to Lady Reading Hospital. Immediate priorities are evacuation of critical zones, medical surge deployment, and clearing road corridors. These figures derive from demonstration data and require field validation before full resource commitment.

Intelligence Assessment

Criticality	critical
Overall Confidence	91%
Escalation Required	True
Rationale	Critical flood emergency in Peshawar with 153 km² inundated, ~540,000 people affected, 14 hospitals at ris
Key Decisions	Authorize immediate evacuation of critical flood zones FZ-01 and FZ-02; Approve identification and clearin
Human Review Required	True

Map Narrative

This static risk map identifies critical flood inundation across 153.37 km² of Peshawar, utilizing SAR imagery to penetrate cloud cover. It highlights 22 high-risk zones where floodwaters intersect with critical infrastructure, indicating a critical vulnerability score of 8.2 and severe threats to healthcare access and transportation corridors.

- 153.37 km² of total inundated area across 22 distinct zones.
- 89 km of blocked road networks restricting emergency mobility.
- High-risk overlap between deep-water zones and 14 healthcare facilities.
- Critical bottleneck identified with only one operational evacuation route for 540,000 affected people.

Priority Timeline

Next 6 Hours	Evacuate high-risk flood zones FZ-01 and FZ-02; Secure access to Lady Reading Hospital; Deploy search
Next 24 Hours	Clear 89km of blocked road corridors; Establish emergency medical surge centers; Open temporary shelte
Next 72 Hours	Restore critical utility services to schools and clinics; Conduct structural assessments of damaged infrastru
Resources	High-clearance vehicles and water rescue boats; Emergency medical supplies and surge staffing; Tempora
Coordination	Multi-agency traffic control for sole evacuation route; Hospital administration for capacity and risk managen

Anomalies and Warnings

- MEDIUM: PDF and map URLs are empty Handling: Verify output generation of visual reports and maps
- HIGH: Critical population exposure (540k) with only one evacuation route Handling: Prioritize immediate identification and clearing of alternative access routes
- GeoJSON artifact URL is not available in local demo mode.
- Map URL is generated later by the local artifact step.
- PDF and map URLs are empty; visual assets missing.
- Single evacuation route creates severe bottleneck for 540,000 exposed residents.
- Impact figures are demonstration-derived and require field validation before deployment.

Quality Check

Status	not_ready
Event Id Present	True
Satellite Data Present	True
Hazard Data Present	True
Impact Data Present	True
Map Artifacts Present	True
Recommendations Present	True
Confidence Above Threshold	True
Warnings	GeoJSON artifact URL is not available in local demo mode.; Map URL is generated later by the local artifac
Blocking Issues	Human validation explicitly required before resource commitment or large-scale response activation.; Critic

Band-Ready Final Message

HazardMind Report Agent complete for demo-peshawar-flood (Peshawar, Pakistan, Flood). Criticality: critical; confidence: 91%. Summary: Peshawar faces a CRITICAL flood emergency with approximately 540,000 people exposed across 153.37 km2 of inundation in 22 zones, confirmed by SAR imagery at high confidence (0.91) despite 42% cloud cover. The crisis threatens 14 hospitals, blocks 89 km of roads, and affects 67 schools, with a vulnerability score of 8.2/10. A severe operational bottleneck exists as only one evacuation route currently serves the entire affected population, while deep-water zones FZ-01 and FZ-02 endanger access to Lady Reading Hospital. Immediate priorities are evacuation of critical zones, medical surge deployment, and clearing road corridors. These figures derive from demonstration data and require field validation before full resource commitment.

Detailed Incident Analysis

Peshawar, Pakistan is experiencing a CRITICAL flood scenario with 153.37 km2 affected and 540,000 people exposed. Flood risk is classified as CRITICAL with hospitals, roads, and schools requiring immediate operational attention.

Technical Analysis

SAR analysis using Sentinel-1 VV/VH ratio (mean 0.24) indicates extensive flooding with 153.37 km2 affected (24.3% of 22 zones). Deep water zone FZ-01 (12.4 km2) and water zone FZ-02 (8.7 km2) dominate critical/high severity areas. Cloud cover (42%) necessitated SAR selection. Impact is severe: 540,000 population affected, 14 hospitals at risk (Lady Reading Hospital high), 89 km roads blocked, 67 schools impacted, and vulnerability score 8.2/10. Only one evacuation route identified. Flood confidence is high (0.91); co-existing earthquake (0.67) and landslide (0.54) risks are moderate to low.

Recommendations

- Prioritize evacuation in critical flood zones.
- Deploy emergency medical support near hospitals at risk.
- Clear blocked road corridors for rescue access.
- Open temporary shelters for displaced families.
- Monitor flood expansion using updated satellite imagery.
- Evacuate high-risk flood zones FZ-01 and FZ-02
- Secure access to Lady Reading Hospital
- Deploy search and rescue to inundated areas
- High-clearance vehicles and water rescue boats
- Emergency medical supplies and surge staffing

Response Priorities

- Activate evacuation support for FZ-01 and FZ-02.
- Protect hospital access and stage medical surge teams near high-risk facilities.
- Clear blocked road corridors needed for rescue and logistics movement.

Assumptions

- Satellite, hazard, and impact values are local demo data until upstream agents publish live outputs.
- Risk zone geometries are simplified mock polygons for dashboard and report demonstration.
- Facility and route coordinates are approximate and intended for prototype visualization.

Limitations

- No live Band messages are consumed in this local demo run.
- No field validation, hydrologic model, or live basemap tiles are used for artifact generation.
- Impact estimates should be treated as decision-support indicators, not official counts.

Model Source Note

Detailed report generated using Featherless Kimi K2.6. Executive summary generated using AI/ML API. Intelligence sources: criticality: featherless:deepseek-v4-pro, anomaly_check: featherless:gemma-4-31b-it, map_narrative: featherless:gemma-4-31b-it, priority_recommendations: featherless:gemma-4-31b-it, decision_brief: aiml:opus-4.8, quality_check: hybrid:deterministic+featherless:qwen-3.6-35b-a3b, band_ready_message: template+intelligence

Detailed Report	featherless:composite
Executive Summary	aiml:opus-4.8
Fallback Used	False
Featherless Model	moonshotai/Kimi-K2.6
Criticality	featherless:deepseek-v4-pro
Map Narrative	featherless:gemma-4-31b-it
Priority Timeline	featherless:gemma-4-31b-it
Quality Check	hybrid:deterministic+featherless:qwen-3.6-35b-a3b

Agent Log / Trace

Agent	Status	Timestamp	Message
hazardmind-report	complete	2026-06-13T18:03:00Z	Report Agent received event context
hazardmind-report	complete	2026-06-13T18:03:20Z	Featherless composite generated detailed disaster report
hazardmind-report	complete	2026-06-13T18:03:40Z	AI/ML generated executive summary
hazardmind-report	complete	2026-06-13T18:03:45Z	Criticality assessed
hazardmind-report	complete	2026-06-13T18:03:50Z	Anomalies checked
hazardmind-report	complete	2026-06-13T18:03:55Z	Map narrative generated

Agent	Status	Timestamp	Message
hazardmind-report	complete	2026-06-13T18:04:00Z	Priority timeline generated
hazardmind-report	complete	2026-06-13T18:04:05Z	Decision brief generated with Opus
hazardmind-report	complete	2026-06-13T18:04:10Z	Quality check completed
hazardmind-report	complete	2026-06-13T18:04:15Z	Band-ready final message prepared
hazardmind-report	complete	2026-06-13T18:04:00Z	Static cartography map generated locally